



LESSON 7.6

INTERPRETING LINEAR RELATIONSHIPS FROM GRAPHS AND EQUATIONS

LESSON INTRODUCTION

MA.8.AR.3.5 Given a real-world context, determine and interpret the slope and y-intercept of a two-variable linear equation from a written description, a table, a graph, or an equation in slope-intercept form.

LEARNING TARGETS

- I can determine and interpret the slope and y-intercept of a two-variable linear equation from a graph in a real-world context.
- I can determine and interpret the slope and y-intercept of a two-variable linear equation from an equation in slope-intercept form in a real-world context.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

MA.7.AR.4.4 Given any representation of a proportional relationship, translate the representation to a written description, table, or equation.

WORKING TOWARD

MA.912.AR.2.5 Solve and graph mathematical and real-world problems that are modeled with linear functions. Interpret key features and determine constraints in terms of the context.

ESSENTIAL QUESTIONS

- What two quantities are being compared in the linear relationship?
- How do you determine and interpret the slope and y-intercept of a two-variable linear equation from a graph and from an equation in slope-intercept form in a real-world context?

LESSON NARRATIVE

Students will determine and interpret the slope and y-intercept in context when given either a graph or an equation in slope-intercept form. Students will also be asked to consider a real-world context where some aspects of the graph do not make sense to the context. This is used to highlight the benefits and shortfalls of mathematical models and how important it is to analyze what the models show to ensure they make sense to the context.

COMPONENT SUMMARY

7.6.1 WARM-UP (~5 MIN.)

Watch a growth mindset video and reflect on how they have grown personally

7.6.3 COLLABORATION (15 MIN.)

Find and interpret the slope and y-intercept and determine if their meanings are reasonable in context

7.6.5 GUIDED PRACTICE (5 MIN.)

Practice determining and interpreting the slope and y-intercept from an equation in context

7.6.7 WRAP-UP (~5 MIN.)

Reflect on their understanding of the lesson's learning targets

7.6.2 GUIDED INSTRUCTION (10 MIN.)

Determine and interpret the slope and y-intercept from graphs and equations in context

7.6.4 BRING IT TOGETHER (5 MIN.)

Teacher debriefs and summarizes learning about mathematical models in context

7.6.6 YOUR TURN! (5 MIN.)

Practice determining and interpreting the slope and y-intercept from a graph in context

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Slope
- Rate of change
- y-intercept

MATERIALS

- Calculators

ADDITIONAL PREPARATION

- Warm-Up 7.6.1 contains a video to accompany the question prompts.
- Bring It Together 7.6.4 contains a video to accompany the question prompts

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not recognize that slope and rate of change are interchangeable.
- Students may struggle with identifying and interpreting the important information from the real-world context of the problem.
- Students may struggle to remember to use both units when interpreting the rate of change in context.

THINGS TO CONSIDER

- This unit provides excellent opportunities to spiral learning from earlier in the unit.
- Allow students to use calculators for this lesson in order to focus on the learning targets of the lesson. Consider requiring students to write the calculations they enter in their calculators to reinforce their computational skills and provide an opportunity for them to check their work.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

Consider implementing a Think-Pair-Share routine for the 7.6.3 Collaboration activity on questions 8 and 9. This strategy will allow students to work with each other and share ideas about whether the key features make sense in the real-world context prior to a whole-class discussion.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD**MA.912.MTR.7.1 Real-World Contexts**

Students will determine the slope and y-intercept from either a graph or an equation and interpret their findings within various real-world contexts given. Asking questions regarding whether the meaning of the slope and y-intercept make sense in each of these contexts will help to build stronger understanding of these relationships in the real-world.

DIFFERENTIATE INSTRUCTION

During 7.6.2 Guided Instruction, remind students how to find the slope using multiple methods (slope triangles for graphs as well as finding the change in y over the change in x). Encourage students to use their preferred method throughout the lesson. Throughout the lesson, encourage students to label, underline, and circle important information found in the model or real-world context. This can be modeled for students during the 7.6.2 Guided Instruction to help those visual learners.

7.6.1 WARM-UP: SUCCESS VS. FAILURE**TEACHER MOVES**

During independent reflection, monitor for student answers that others would benefit from hearing. Probe students for who is willing to share what they have done to improve, and challenge them to make connections on how they can apply these methods to other areas or their lives.

**7.6.1 Warm-Up: Success vs. Failure**

John Legend shares that even though he is seen as a successful musician, he actually tried to get a record deal for six years without any success. He explains that most successful people go through a lot of missteps before reaching their goal.

He said, "The great ones aren't the ones who don't fail, they're the ones who learn from their failure, and they keep going."



John Legend explains that innate talent is not always enough. He only went to the next level in his career after hiring a voice coach and studying to get better.

Think of something in your life you are successful with.

1. What have you done to grow and improve this talent?
Answers may vary.



7.6.2 Guided Instruction: Interpreting Slope and y-intercept from Equations and Graphs

When linear relationships represent real-world contexts, the slope and y-intercept of the relationship have meaning within the context.

- The ☐ slope ☐ y-intercept represents the ☐ initial value, ☐ rate of change, or how one quantity changes in relationship to the other.
The ☐ slope ☐ y-intercept represents the ☐ initial value, ☐ rate of change, or the value of y when $x = 0$.

- Lyla notices that the level of the water in her swimming pool drops each week if she doesn't regularly refill it, especially in the hot summer months. The equation $y = -1.6x + 42$ represents the height of the water at the shallow end of the pool, y , in inches after x number of weeks beginning in June.
 - Which two quantities are being compared in this linear relationship?
Water height, in inches, and time, in weeks.
 - Use the equation to determine the slope and the y-intercept in this relationship.

Slope	y-intercept
-1.6	42

- Complete each statement to interpret the meaning of each value in context.

The slope means ☐ the height of the water in the pool ☐ the number of weeks since June 1st
is ☐ increasing ☐ decreasing by ☐ 1.6 weeks ☐ 1.6 inches ☐ 42 weeks ☐ 42 inches each ☐ inch ☐ week beginning in June.

The y-intercept means ☐ the height of the water in the pool ☐ the number of weeks since June 1st
is ☐ 1.6 weeks ☐ 1.6 inches ☐ 42 weeks ☐ 42 inches on June 1st.

7.6.2 GUIDED INSTRUCTION: INTERPRETING SLOPE AND Y-INTERCEPT FROM EQUATIONS AND GRAPHS

TEACHER MOVES

During whole-group instruction, use modeling questioning to scaffold students making connections between their knowledge of finding the slope and y-intercept and how to interpret their meanings in context.

ENRICHMENT

How does the sign of the slope relate to its interpretation in context?

DIFFERENTIATE INSTRUCTION

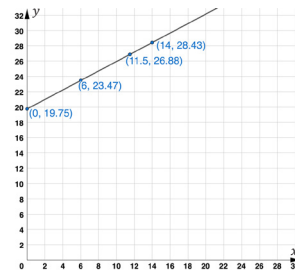
Consider allowing students to complete the questions individually or with a partner. Debrief the questions with a whole-class discussion to solidify student understanding.

7.6.2 GUIDED INSTRUCTION: INTERPRETING SLOPE AND Y-INTERCEPT FROM EQUATIONS AND GRAPHS (CONTINUED)

ENRICHMENT

- Which points did you use to determine the slope?
- Is there more than one way to determine the slope of the line?
- Is there more than one way to determine the y-intercept from the information given?

3. The graph models the relationship between the average height of a male infant in inches, y , from birth to two years old, where x is the number of months since birth.



Use the graph to determine the rate of change and y-intercept. Then, interpret each value in context.

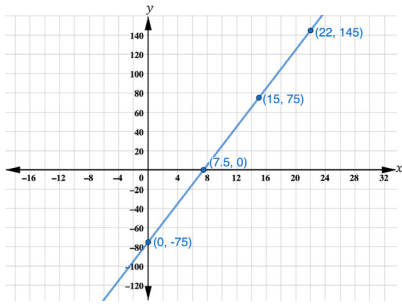
$$\frac{23.47 - 19.75}{6 - 0} = \frac{3.72}{6} = 0.62$$

Feature	Value	Interpretation
rate of change	0.62	The average male infant's height increases 0.62 inches per month.
y-intercept	19.75	The average height of a male infant at birth is 19.75 inches.



7.6.3 Collaboration: Making Sense of Mathematical Models

The Ridgeview High School cross-country team ran a car wash on Saturday to raise money for their team’s new uniforms. The graph represents the relationship between the amount of money the team raised to buy uniforms, y , and the number of cars they washed, x .



Work with your partner to answer the questions using the information given in the graph.

- 1. What is the slope of the line? Show your work.
$$\frac{75-0}{15-7.5} = \frac{75}{7.5} = 10$$
- 2. What does the slope mean in this context?
They raised \$10 for every car they washed.
- 3. Explain whether the meaning of the slope makes sense in this context.
It is reasonable that the team charged \$10 per car, so the slope makes sense in context.

- 4. What is the y -intercept of the line?
 -75
- 5. What does the y -intercept mean in this context?
Before the team washed any cars, they had $-\$75$ raised to buy uniforms.
- 6. Explain whether the meaning of the y -intercept makes sense in this context.
The y -intercept makes sense in this context because it’s likely the team had \$75 worth of expenses before they started the car wash in buying supplies needed.

7. Complete the table.

Point	What does this point represent in this context?	Explain whether the meaning of the point makes sense in this context.
(7.5, 0)	<i>After washing 7.5 cars, the team has raised \$0 to spend on uniforms.</i>	<i>It doesn’t make sense in context because they wouldn’t wash $\frac{1}{2}$ a car.</i>
(22, 145)	<i>The team has raised \$145 for uniforms after washing 22 cars.</i>	<i>It makes sense in this context.</i>

- 8. What parts of the graph do not make sense in this context?
Fractional, decimal, or negative values of x , the number of car washes
- 9. Discuss with your partner whether the graph is useful for modeling this situation, even if there are parts of it that do not make sense given the context.

7.6.3 COLLABORATION: MAKING SENSE OF MATHEMATICAL MODELS

TEACHER MOVES

Listen to students’ discussions of the meaning of the slope and whether it makes sense in the context. Students should include both units when interpreting the slope.

ENRICHMENT

What does a negative y -value mean in this context?

TEACHER MOVES

Consider having a whole-group discussion addressing questions 8 and 9. Monitor the partner work to find student ideas to share with the class and then encourage students to share their thinking during the whole-group discussion.

7.6.4 BRING IT TOGETHER: MAKING SENSE OF MATHEMATICAL MODELS

QUESTIONING

If time allows, have students actually consider these questions with partners, using the graph.



7.6.4 Bring it Together: Making Sense of Mathematical Models

Mathematical modeling is a process that uses mathematics to represent, analyze, and make predictions about real-world phenomena.

Today's lesson explores two mathematical models – equations and graphs.

Mathematical models are helpful tools to use to understand a situation and make predictions. However, sometimes the mathematical model can include parts that do not make sense in a real-world context.

In the car wash example, washing part of a car or a negative number of cars does not make sense, but the model can still be a useful tool for answering questions such as, "After how many car washes will the team have covered their expenses?" or "After how many car washes will the team have raised \$100?"

When interpreting models, it is important to analyze whether aspects of the model make sense given the context.



7.6.5 Guided Practice: Interpreting Slope and y-intercept in Real-World Contexts

- Mila is doing an experiment where she fills a container with water a little bit at a time. Each time she adds water to the container, she measures both the height of the water, h , in inches and the weight of the container, w , in ounces. The relationship she observes can be modeled with the equation $w = 12.4h + 2.5$.

Use the equation to determine the rate of change and y-intercept. Then, interpret each value in context.

Feature	Value	Interpretation
rate of change	12.4	<i>The weight of the container increases 12.4 ounces each time the water's height increases by 1 inch.</i>
y-intercept	2.5	<i>The container weighed 2.5 ounces when it was empty.</i>

7.6.5 GUIDED PRACTICE: INTERPRETING SLOPE AND Y-INTERCEPT IN REAL-WORLD CONTEXTS

DIFFERENTIATE INSTRUCTION

Stress the importance of units in a real-world context. Encourage students who are struggling to circle or underline the units to help them interpret the rate of change.

Consider having students summarize the situation in their own words with a partner after reading the prompt to ensure understanding of the context.

TEACHER MOVES

Consider making this a Think-Pair-Share activity and partner students accordingly to encourage discussions to help students make deeper connections in their learning.

7.6.6 YOUR TURN!

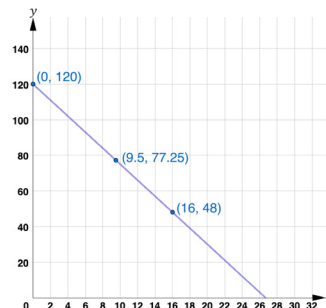
ENRICHMENT

- What are the quantities being compared in this context?
- Which points did you use to find the slope?
- How can you determine the y-intercept from the graph? From the context?



7.6.6 Your Turn!

1. Marcus collects rainwater in a 120-gallon barrel to use for watering the plants in his garden during periods of time when it does not rain as much. The graph models the relationship between the number of gallons in the barrel, y , and the number of days without rain, x .



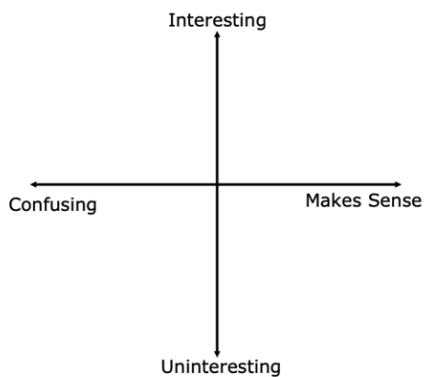
Use the graph to determine the slope and y-intercept. Then interpret each value in context.

$$\frac{48-120}{16-0} = \frac{-72}{16} = -4.5$$

Feature	Value	Interpretation
slope	-4.5	The amount of water in the barrel decreases by 4.5 gallons of water each day without rain.
y-intercept	120	There were 120 gallons of water in the barrel before any was used to water plants.

**7.6.7 Wrap-Up: Reflection**

1. Plot a point on the grid to indicate your level of interest and understanding for today's learning targets.



2. Explain why you placed the point where you did.
Answers will vary.

7.6.7 WRAP-UP: REFLECTION**TEACHER MOVES**

Consider discussing anything that the students still find confusing and make a list of the topics on the board or a poster to come back and address later.